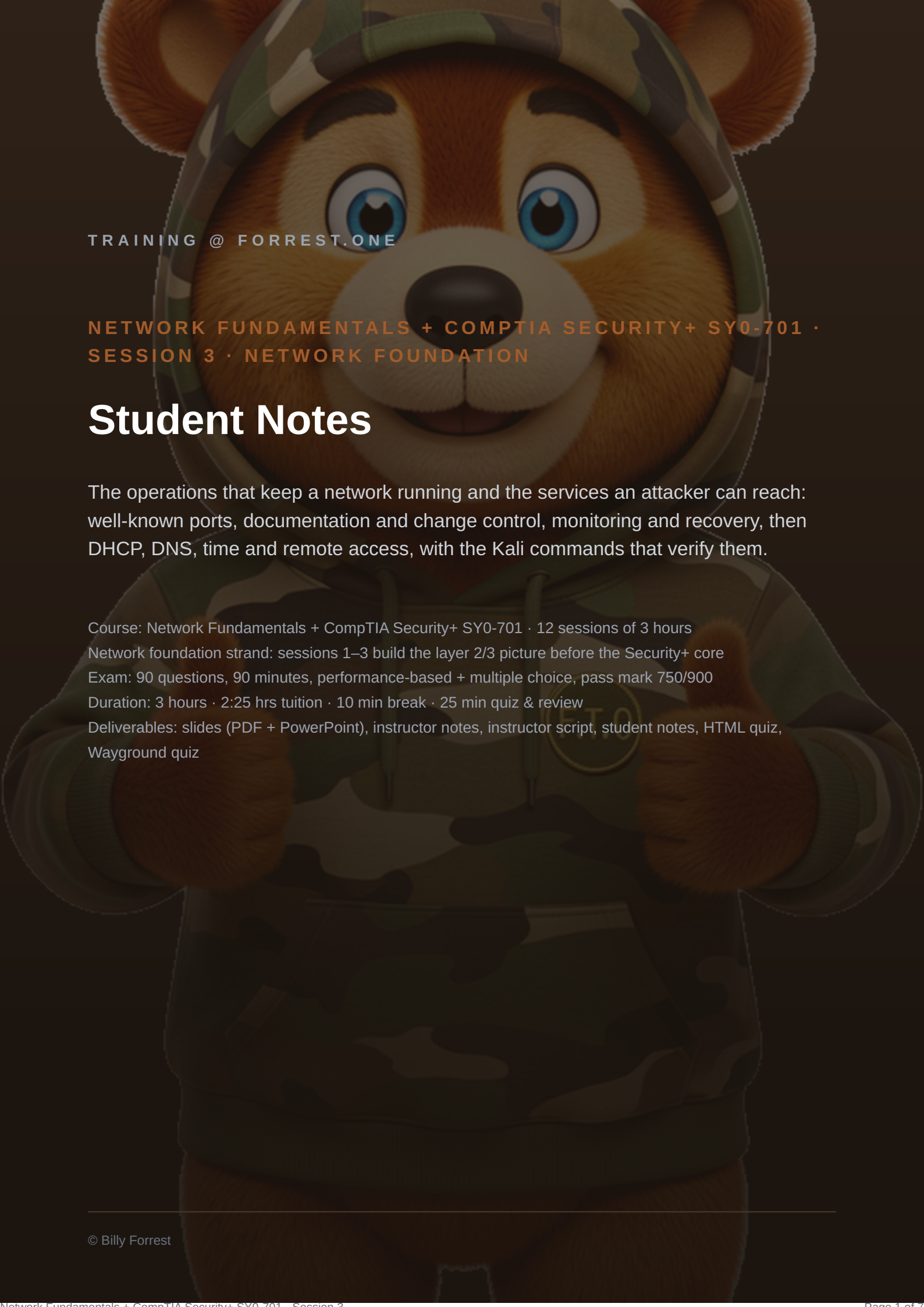




TRAINING @ FORREST.ONE

NETWORK FUNDAMENTALS + COMPTIA SECURITY+ SY0-701 ·
SESSION 3 · NETWORK FOUNDATION

Student Notes

The operations that keep a network running and the services an attacker can reach: well-known ports, documentation and change control, monitoring and recovery, then DHCP, DNS, time and remote access, with the Kali commands that verify them.

Course: Network Fundamentals + CompTIA Security+ SY0-701 · 12 sessions of 3 hours

Network foundation strand: sessions 1–3 build the layer 2/3 picture before the Security+ core

Exam: 90 questions, 90 minutes, performance-based + multiple choice, pass mark 750/900

Duration: 3 hours · 2:25 hrs tuition · 10 min break · 25 min quiz & review

Deliverables: slides (PDF + PowerPoint), instructor notes, instructor script, student notes, HTML quiz, Wayground quiz

Student Notes, Network Services, Operations & Troubleshooting

Your revision guide for Session 3, the final session of the network-foundation strand. It follows the six chapters of the session: ports and traffic types, documentation and agreements, life cycle and change, configuration and monitoring, baselines, logs and resilience, and the core services with remote access. Read alongside the slides, run the commands on any Linux box you can reach, then use the self-check questions and the session quiz to confirm you are ready for the Security+ core.

Key terms

Term	Definition
Port	A numbered endpoint for a service; every firewall rule and troubleshooting step starts with one
FTP / SFTP	File transfer in plaintext on 21 and 20; Secure File Transfer inside SSH on 22
Telnet / SSH	Plaintext remote shell on 23; encrypted remote shell on 22
SMTP	Mail relay between servers on 25; client submission on 587
HTTP / HTTPS	Web on 80; web inside TLS on 443
SNMP	Device management: 161 for requests, 162 for traps
LDAP / LDAPS	Directory queries on 389; encrypted directory queries on 636
SIP	Call signalling on 5060; encrypted signalling on 5061
Unicast	One sender to one receiver
Broadcast	One sender to every device on the segment; stays inside one subnet
Multicast	One sender to the devices that joined the group
Anycast	One sender to the nearest of many servers sharing one address
Physical diagram	Devices, ports in use and cable runs: where the cable goes
Logical diagram	Addressing, VLAN segments and traffic flow: how traffic moves
Rack unit	1.75 inches of vertical space in a cabinet
IPAM	IP address management: tracks static and dynamic allocation and finds conflicts
SLA	Service level agreement: uptime, response time and resolution time

Term	Definition
The nines	Each additional nine cuts permitted downtime roughly tenfold
End of Life	The vendor stops selling the product
End of Support	No further patches; every flaw found afterwards stays open
Configuration drift	Slow divergence of a running device from its approved baseline
SNMP trap	An unsolicited alert sent from a device to the manager on port 162
Community string	A plaintext password in older SNMP versions; version 3 adds authentication and encryption
Flow record	A summary of a conversation: addresses, ports, protocol, bytes, times
NetFlow / sFlow / IPFIX	Flow collection as traffic crosses an interface, packet sampling, and the standard export format
Full packet capture	Every byte of every packet; highest detail, highest storage cost
Baseline	What normal looks like, recorded: bandwidth, latency, error rate and load
Syslog severity	Zero is emergency, seven is debug
SIEM	Correlates logs across sources to surface a pattern no single device shows
RPO	Recovery Point Objective: how much data you may lose; drives backup frequency
RTO	Recovery Time Objective: how long you may be down; drives failover
MTTR / MTBF	Mean Time To Repair and Mean Time Between Failures
Cold, warm, hot site	Space only; some equipment; a live mirror, from cheapest and slowest to costliest and fastest
Active-active	Every node serves traffic and shares the load
Active-passive	A standby takes over when a heartbeat stops
DORA	Discover, Offer, Request, Ack: the DHCP lease exchange
Relay agent	Carries DHCP conversation to a server on another subnet
Reservation	Pins one address to one hardware address
SLAAC	Stateless Address Autoconfiguration: the host builds its address from a router advertisement
DHCPv6	Stateful assigns and records addresses; stateless supplies settings only
Privacy extensions	The host rotates temporary IPv6 addresses, resisting long-term tracking
TTL	Time to live: how long a resolver may cache a DNS answer
Split horizon DNS	Answers internal and external clients differently
DNSSEC	Cryptographic signing of DNS records, so a forged answer fails validation
Stratum	Distance from the authoritative time source: zero is the reference clock

Term	Definition
VPN	Site to site joins two offices; remote access serves one user
IPsec transport / tunnel mode	Encrypts the payload only, or the whole packet
Split tunnelling	Sends only corporate traffic through the tunnel
Console cable	Out-of-band serial management when no network path answers
Attack surface	The sum of entry points an attacker can use to enter or extract data
Threat vector	The path or method an attacker uses to reach the attack surface

Core concepts

Chapter 1: Ports, protocols and traffic types

Learn the ports in pairs, plaintext against encrypted: 21/20 against 22, 23 against 22, 80 against 443, 389 against 636, 5060 against 5061. The protected form always moves to a different port. Rule of thumb: if the name has no S, assume plaintext. Troubleshooting rule: names resolve first, so check port 53 early. Traffic is unicast (one to one), broadcast (one to all on the segment, contained within one subnet), multicast (one to the group that joined) or anycast (one to the nearest of many sharing one address, which is how global name servers and content networks scale).

Chapter 2: Documentation, agreements and surveys

The physical diagram answers where the cable goes; the logical diagram answers how the traffic moves. Both must be current. Rack diagrams fix unit positions and power feeds, cable maps trace paths and termination points, wiring schematics document structured cabling and patch panels, and floor plans mark equipment rooms, cable trays and access points. The asset inventory carries serial number, location, configuration and owner; software entries carry version, licence key and location; warranties carry end dates. Address management tracks every allocation, detects conflicts and feeds changes into the name system. An SLA defines uptime, response time and resolution time, and each additional nine cuts permitted downtime roughly tenfold, which is a design instruction disguised as a contract term. Wireless surveys are predictive before installation and validation afterwards, and the lower band offers only three non-overlapping channels: 1, 6 and 11.

Chapter 3: Life cycle and controlled change

End of Life is a sales decision; End of Support is a security decision, because no further patches arrive and every flaw found afterwards stays open. Update as a process: download, verify the source, stage, test, deploy, with a tested rollback plan in a maintenance window. Decommissioning is a security event: sanitise the configuration, remove certificates and stored credentials, then destroy, degauss or overwrite the media, recycle it under electronic waste rules, and update the inventory, diagrams and monitoring. Change control counters self-inflicted outage: the Change Advisory Board reviews disruptive work, standard changes are pre-approved, normal changes are planned and tested, emergency changes go in first and are reviewed

afterwards, and every request carries justification, risk, impact, rollback and result. Every change touches four records: the diagram, the inventory, the configuration baseline and the monitoring.

Chapter 4: Configuration and monitoring tools

The production configuration is what a device runs now; the baseline configuration is the approved template. Drift is the slow gap between them, found by compliance checking and closed by restoring the baseline. Back up locally, in version control and off site, and test a restore. SNMP has a manager, an agent and a management information base; object identifiers name each value, GET reads and SET changes, and a trap is an unsolicited alarm sent to port 162. A community string is a plaintext password, so use version 3. Flow data summarises conversations: NetFlow collects as traffic crosses an interface, sFlow samples packets, IPFIX standardises the export. Flows are cheap and retain for months; full capture reads what was said, costs storage, and needs authority because packets hold personal data.

Chapter 5: Baselines, logs and resilience

A baseline records bandwidth, latency, error rate and device load under normal conditions, so that a number later means something. Threshold alerting fires on fixed limits; anomaly detection watches the shape of the data; trending forecasts capacity, and a baseline is only valid until the network changes. Log aggregation gives one searchable repository, syslog carries source, timestamp and severity from zero to seven, and a SIEM correlates across sources: one device shows an event, correlation shows the campaign. RPO says how much data you may lose and drives backup frequency; RTO says how long you may be down and drives failover; MTTR and MTBF measure repair speed and component life. Cold, warm and hot sites trade cost against recovery speed. Active-active shares load, active-passive waits on a heartbeat, load balancing steers around dead nodes, and a plan that has never been tested is unproven: use tabletop, simulation, parallel test and full failover, at least once a year.

Chapter 6: Core services and remote access

DHCP lease negotiation is DORA: Discover, Offer, Request, Ack, from no address at all to the broadcast address. A relay agent carries the conversation to another subnet, a reservation pins an address to a hardware address, scope options deliver gateway, name servers and domain, and two servers in failover share the pool. The security catch: a client accepts the first offer it receives, so an unauthorised server on the LAN wins the race and becomes your gateway. IPv6 changes who configures addresses: stateless autoconfiguration lets the host build its own address from a router advertisement, stateful DHCPv6 assigns and records, stateless DHCPv6 supplies options alone, and privacy extensions rotate temporary addresses. DNS runs on root servers, top-level domains and authoritative servers, with resolvers asking on your behalf; records are A, AAAA, CNAME, MX, PTR and TXT, each with a time to live that decides cache lifetime; split horizon answers internal and external clients differently, and DNSSEC signs records so forgery fails. Time is organised in strata from the reference clock downward, needs several sources for cross checking, and must be authenticated; drift breaks tickets, logs and timelines. VPNs are site to site or remote access, IPsec transport mode protects the payload and tunnel mode the whole packet, and split tunnelling sends only corporate traffic. Device management paths are the console cable, Secure Shell, the web interface and the programmatic interface, and every one of them is an attack surface.

Practical cautions

- Do not leave a plaintext service running where an encrypted replacement exists. Telnet on 23 and FTP on 21 are the usual offenders.
- Do not run SNMP with a community string on a reachable interface. Use version 3, and filter port 161 to management hosts.
- Do not trust a diagram or inventory you did not update at the last change. A wrong drawing is worse than no drawing.
- Do not let equipment run past End of Support without a compensating control and a replacement date.
- Do not retire a device with its configuration, certificates or credentials still on it.
- Do not make a change without a rollback plan or outside a maintenance window.
- Do not assume a device matches its baseline. Check for drift, and back the configuration up off the device.
- Do not capture packets without authority: they contain personal data.
- Do not buy failover before you have agreed the RPO and RTO with the business.
- Do not assume a hot site removes the need to test. An untested plan is still unproven.
- Do not leave DHCP unguarded: a client accepts the first offer it receives.
- Do not trust a forensic timeline from a system whose clock was never verified against an authenticated source.
- Do not leave an unexpected listener running. Explain every entry in `ss -tuln`, or treat it as a finding.

Self-check questions

Answers are in your session slides

If you cannot answer a self-check from memory, revisit the corresponding slide, then retake the quiz.

Exam tips for Session 3

- The port table is core knowledge: 20/21 FTP, 22 SSH and SFTP, 23 Telnet, 25/587 SMTP, 53 DNS, 67/68 DHCP, 80/443 web, 161/162 SNMP, 389/636 LDAP, 3389 RDP, 5060/5061 SIP.
- Encrypted services usually move to a different port. If a question gives you an unfamiliar port, ask which service moved there.
- Traffic types are examinable: a broadcast stays within one subnet, a multicast reaches only group members, and anycast reaches the nearest of many.
- End of Support is the security cliff. A question about unsupported equipment is a question about unpatched vulnerabilities.
- RPO is about data loss and backup frequency; RTO is about downtime and failover. Do not swap them.
- Match the control to the attack: documentation and baselines against drift, change control against self-inflicted outage, DNSSEC against DNS forgery, DHCP snooping against rogue servers.

- Flows tell you what happened, packets tell you what was said. A question about retention or storage cost is a flow question.
- A plan that has never been tested is unproven: tabletop, simulation, parallel, then full failover, in increasing order of proof.
- Every management path is attack surface: console, command line, browser and programmatic interface.
- Reducing the attack surface is a stated security principle, and closing unused services is the example.

Further reading

- Internet Assigned Numbers Authority service name and port number registry.
- RFC 2131 and RFC 8415: DHCP and DHCPv6.
- RFC 1034 and RFC 1035: DNS concepts and implementation. RFC 4033: DNSSEC introduction.
- RFC 5905: NTP version 4. RFC 8915: Network Time Security.
- RFC 4251: SSH protocol architecture.
- ITIL 4 change enablement guidance, for the change control vocabulary.
- UK National Cyber Security Centre: network security and logging guidance.